

Bullwinkle

Q1.

Overhead allocation rate:

Overhead	Practical Capacity
\$100,000	1,000

Allocation rate = \$100 per unit

2023:

COGS =	600 units @ \$100 = \$60,000
Cost of capacity (expense)*	\$40,000
Total Expense/COGS	\$100,000

2024:

COGS =	750 units @ \$100 = \$75,000
Cost of capacity (expense)*	\$25,000
Total Expense/COGS	\$100,000

*This number is calculated as a "plug" (the amount necessary so the numbers sum to \$100,000).

Q2

2023:

Year-end Finished Goods Inventory: 100 units @ \$100 = \$10,000

COGS =	500 units @ \$100 = \$50,000
Cost of capacity (expense)*	\$40,000
Total Expense/COGS	\$90,000

Total allocated \$100,000

Q2A

2023:

Year-end Finished Goods Inventory = 200 units @ \$100 = \$20,000

COGS =	500 units @ \$100 = \$50,000
Cost of capacity (expense)*	\$30,000
Total Expense/COGS	\$80,000

Total allocated \$100,000

Note that total expense/COGS goes down relative to previous case, because some of the overhead cost is now allocated to finished goods inventory on the balance sheet; this means the company reports higher gross margin. This is one lever managers can use to report higher margins.

Q3

Overhead allocation rate:

Overhead	Practical Capacity
\$100,000	900

Allocation rate = \$111 per unit

2023:

Year-end Finished Goods Inventory = 100 units @ \$111.11 = \$11,111

COGS =	500 units @ \$111.11 = \$55,555
Cost of capacity (expense)*	\$33,334
Total Expense/COGS	\$88,889

Total allocated	\$100,000
-----------------	-----------

Here again, COGS goes down relative to the previous case, because more of the overhead is allocated to finished goods inventory, and so again shows higher gross margin. But here it's due to a lower assumed practical capacity, which is another lever managers can use.

Q4

The difference with variable manufacturing overhead is that we use actual production, not practical capacity, as the denominator in the allocation calculation, which means the per unit cost allocation changes with as the level of production changes.

Overhead allocation rate for 2023:

Overhead	Actual Production
\$100,000	600

Allocation rate = \$167 per unit

COGS =	600 units @ \$166.67 = \$100,000
Cost of capacity (expense)	\$0
Total Expense/COGS	\$100,000

Overhead allocation rate for 2024:

Overhead	Actual Production
\$100,000	750

Allocation rate = \$133 per unit

COGS =	750 units @ \$133.33 = \$100,000
Cost of capacity (expense)	\$0
Total Expense/COGS	\$100,000

Note that the company could still increase production artificially to increase inventory on hand at the end of the period, which would reduce reported COGS: the more units produced, the lower the per-unit \$ of overhead allocated to COGS.

For example, if in 2023 production was increased to 700 units, the per unit cost allocation would decline to \$142.86 and COGS would therefore decline to \$85,716, increasing the reported gross margin (still selling 600 units so Revenue doesn't change).